

Lesson Plan:

1. Introduction to Matryoshka Dolls:

Show students a set of Matryoshka dolls, each fitting inside a larger one.

Explain that every doll is similar in shape and appearance, but smaller ones are enclosed within bigger ones.

2. Connecting to C++ Nested Loops:

Explain that in C++, a nested loop is inside another loop.

Just like smaller dolls are placed inside larger ones, an inner loop runs completely for every single iteration of the outer loop.

3. Example Code:

```
for (int i = 1; i <= 3; i++) { // Outer loop (larger doll)
    for (int j = 1; j <= 2; j++) { // Inner loop (smaller doll)
        std::cout << "Outer: " << i << ", Inner: " << j << "\n";
    }
}
```

Each outer iteration contains multiple inner iterations like each large doll contains smaller dolls.

4. Conclusion:

The nesting structure in Matryoshka dolls visually represents how loops work in layers.

This concept helps in tasks like printing patterns, handling multi-dimensional arrays, and solving complex problems efficiently.